RDMS Using ORACLE-2019 New

Q1-A

- (1). Row types are used to store which type of data ?
  Ans. Multiple data values
- (2). A table can't have one more than one long data type (True/False).
  Ans. ________________ .
- (3). ________removes all the rows from table, but its structure and constraints and so on, remains.
  Ans. TRUNCATE TABLE statement
- (4). In SQL, which command is used to select only one copy of each set of duplicate rows ?
  Ans. DISTINCT COMMAND

Q1-B

1. Write a note on E-R Diagram.

- E-R Diagram stands for Entity Relationship Diagram.
- E-R Diagram is a diagram that represents relationships among entities in a data
- **Rectangles** represent table sets.
- **Diamonds** represent relationship between tables.
- **Lines** link attributes represent connection between tables with relationship and field.
- **Ellipses** represent table fields
- **Underline** indicates primary key attributes

2. Differentiate: DBMS V/S RDBMS.

DBMS	RDBMS
• RELATIONSHIP BETWEEN TWO TABLES OR FILES ARE MAINTAINED PROGRAMICALLY	• RELATIONSHIP BETWEEN TWO TABLES OR FILE CAN BE SPECIFIED AT THE TIME OF TABLE CREATION
• DOES NOT SUPPORT CLIENT /SERVER ARCHITECTURE	• SUPPORTS CLIENT/SERVER ARCHITECTURE
• DOES NOT SUPPORT DISTRIBUTED DATABSES	• SUPPORTS DISTRIBUTED DATABASES
• NO SECURITY OF DATA	• MULTIPLE SECURITY LEVES; 1 LOGIN AT OS LEVEL 2 COMMAND LEVEL SECURITY 3 OBJECT LEVEL SECURITY
• EACH TABLE IS GIVEN EXTENSION	• MANY TABLES ARE GROUPED IN ONE DATABASE IN RDBMS

Q1-C

1.Explain various components of sql.

- **There are four types of SQL command.**

DDL (Data Definition Language)
DML (Data Manipulation Language)
DCL (Data Control Language)
DQL (Data Query Language)

- **DDL (Data Definition Language)**
- **CREATE:** It is used to create objects (table, view, function, procedure, trigger etc.) in the database.
- **ALTER:** It is used to change the structure of database.
- **DROP:** It is used to delete objects from the database.
- **TRUNCATE:** It is used to remove all records from the table, including all spaces allocated for the records are removed.
- **GRANT:** It is used to give different command grant (access rights) to the user.
- **REVOKE:** It is used to take the given grant back from the user.
- **DML (Data Manipulation Language)**
- **INSERT:** it used to insert the data into a table
- **UPDATE:** it used to update the existing data within a table.
- **DELETE**: it used to delete all the records from the table, the space will be remaining.
- **CALL:** it is used to call a PL/SQL program.
- **DCL (Data Control Language)**
- **COMMIT:** it is used to save the work.
- **SAVEPOINT:** it is used to identify a point in transaction.
- **ROLLBACK:** Restore database to original since the last commit (like undo).
- **DQL (Data Query Language)**
- **SELECT:** It is used to retrieve data from the database.

2. What are E-F code rules? Explain.

- Rule 1: The Information Rule – All data should be presented in table form.
- Rule 2: Guaranteed Access Rule – All data should be accessible without ambiguity. This can be accomplished through a combination of the table name, primary key and column name.
- Rule 3: Systematic Treatment of Null Values –A field should be allowed to remain empty. This involves the support of a null value, which is distinct from an empty string or a number with a value of zero.

• Rule 4: Dynamic On-Line Catalog based on the Relational Model –A relational database must provide access to its structure through the same tools that are used t access the data. This is usually accomplished by storing the structure definition within special system tables.
• Rule 5: Comprehensive Data Sublanguage Rule –The database must support at least one clearly defined language that includes functionality for data definition, data manipulation, data integrity, and database transaction control.
• Rule 6: View Updating Rule –Data can be presented in different logical combinations called views. Each view should support the same full range of data manipulation that has direct access to a table available. In practice, providing update and delete access to logical views is difficult and is not fully supported by any current database.

Q1-D

1. List out various operators and explain any 2 operators with example.

- An operator is a symbol that tells the compiler to perform specific mathematical or logical manipulation.
- PL/SQL language is rich in built-in operators and provides the following types of operators
- **Comparison operators**
- **Arithmetic operators**
- **Relational operators**
- **Logical operators**

- **Types Of Logical operators:**

Operator	Description	Examples
• and	• Called the logical AND operator. If both the operands are true then condition becomes true.	• (A and B) is false.
• or	• Called the logical OR Operator. If any of the two operands is true then condition becomes true.	• (A or B) is true.
• not	• Called the logical NOT Operator. Used to reverse the logical state of its operand. If a condition is true then Logical NOT operator will make it false.	• not (A and B) is true.

- Types of Comparison operators
    - IN
    - Between
    - Like
- The IN operator is used when you want to compare a column with more than one value. It is similar to an OR condition.
- Example
    - select *From emp
    - where deptno in(10,20);
- The operator BETWEEN and AND, are used to compare data for a range of values.
- This operator also known as range operator.
- Example
    - select *From emp
    - where sal between 800 and 1500;
- This operator is known as Pattern Matching operator
- The LIKE operator is used to list all rows in a table whose column values match a specified pattern.
- Example
    - SELECT *FROM EMP
    - WHERE ENAME LIKE 'S%';

3. Explain various data type available in SQL.

- **char**
- size 1 to maximum 2000 characters and if data is short than its specified than space padded rest of size
- **Varchar**
- **Varchar2**
- Both varchar and varchar2 are same, it stores alphanumeric data and also store data in its size if data is short than is length than varchar2 remove extra length of data type and allocates to other data

- **Long**
- Long data type is Large Object (LOB) data types. It stored up to 2GB length. But restriction of this type is you cannot use WHERE clause, Group By clause, Order By clause, Distinct operator in SELECT statement
- **Number**
- This data type used to store negative and positive integers: fixed-point and floating numbers with precisions of up to 38 digits.
- **Date**
- The date data type is used to store date and time information. Default date format is DD-MON-YY.

Q2-A

- 1. Clause acts like a where clause but is used for groups rather than rows.

  Ans.HAVING
- 2.The condition in a where clause can refer to only one value. (True / False)

  Ans.False
- 3.Drop table cannot be used to drop a table referenced by a constraints.

  Ans.Foreign key
- 4.The statement in SQL which allows changing the definition of a table is .

  Ans.Alter table

Q2-B

1. Explain group by and having clause with suitable example.

- The group by statement groups rows that have the same values into summary rows, like “find the maxsal from each job in emp table”.
- The group by statement is often used with aggregate functions (count, max, sum, avg) to group the result-set by one more columns.
- By default, group by sorts the rows in ascending order based on the values in the group column

- Group by syntax:

  select <column_name> from <table_name>

  group by <column_name> order by <column_name>;

- Having clause special use for give condition to group by like where.
- You place the having clause after your group by clause.
- Having syntax: select <column_name> from <table_name> group by <column_name> having <condition> order by <column_name>;
- Example:

  Select job, count(sal) group by job having='clerck';

2. Write a note on set operators.

- The SQL set operation is used to combine the two or more SQL SELECT statements.
- **Types of set operation**
  - **Union**
  - **Union all**
  - **Intersect**
  - **Minus**
- **Union**
- The sql union operation is used to combine the result of two or mare sql select queries.
- The union operation eliminates the duplicate rows from its result set.
- Syntax:

  select <column_name> from <table1>

  union

  select <column_name> from <table2>;

- **Union all**
- Union all operation is equal to the union operation. It returns the without removing duplication and sorting the data.
- Syntax:
- select <column_name> from <table1>
- Union all
- Select <column_name> from <table2>;

- **Intersect**
- It is used to combine two select statements. The intersect operation returns the common rows from both the select statements.
- Syntax: select <column_name> from <table1>
- intersect
- select <column_name> from <table2>;
- **Minus**
- It combines the result of two select statements. Minus operator is used to display the rows which are present in the first query but absent in the second query.
- Syntax:
- Select <column_name> from <table1>
- Minus
- Select <column_name> from <table2>;

Q2-C

1. What is join ? Explain its various types.

- A JOIN clause is used to combine rows from two or more tables, based on a related column between them.
- **Different types of sql joins**
  - Inner join
  - Outer join
    - Left outer join
    - Right outer join
    - Full outer join
  - Self join
  - Cross join
- **Inner join:**
- SQL INNER JOIN keyword
- The inner join keyword select records that have matching values in both tables.
- This join is also known as equi join.

- **Outer join**
- SQL left join keyword:
- The left join keyword returns all records from the left table (table1), and the matched records from the right table (table2).
- SQL right join keyword:
- The right join keyword returns all records from the right table (table2), and the matched records from the left table (table1).
- SQL full outer join keyword:
- The full outer join keyword return all records when there is a match in either left (table1) or right (table2) table records.
- **Self join**
- A self join is a regular join, but the table is joined with itself.
- **Cross join**
- In this join there is no condition given so records multiply by each other table
- Like if one table have 3 records and other table have 4 and we apply cross join than result will be 12 rows display.

2. What are constraints ? Explain any two constraints with suitable example.

- Constraints are set of rules used to enforce data integrity.
- They don't take space in database like data.
- Even if single column fails to satisfy constrain, entire record will be rejected by Oracle.
- **Type of constraints are:-**

➢ Primary key
➢ Foreign key
➢ Unique
➢ Not null
➢ Check

- **Unique**
- The purpose of UNIQUE constraint is to ensure that the data in the column is unique and it must not be repeated across the column.
- A table can have more than one unique key column.
- Unique constraint at column level:
- Syntax: <column name data type (size)> unique,
- Unique constraint at table level:
- Syntax: unique(columnname[,columnname, ….])
- **Check**
- Business rule validations can be applied to a column using check constraint.
- Check constraint must be specified as logical expression that evaluates either to true or false.
- Check constraint at column level:
- Syntax: <column name data type (size)> check (logical expression)
- Check constraint at table level:
- Syntax: check(logical expression)

Q2-D

1. **Consider these following tables and only solve the query.**
   Tables:1. student: s_id (Primary key), sname, city
   2. result: r_id (Primary key), mark1, mark2, s_id (foreign key)

- Query:-
- Write a SQL statement to make a list with r_id, s_id, student name and mark1 for those whose marks between 70 and 90.
- Find out maximum mark2 from result table.
- Ans.
- Create table student

  (s_id number(3) primary key,

  Sname varchar2 (30),

  City varchar2 (20));

- Create table result

    (r_id number(3) primary key,

    Mark1 number(3),

    Mark2 number(3),

    S_id number foreign key (s_id) references student (s_id));

- Select s.sname, r.mark1 from student s, result r

  where s_id = s_id, between 70 and 90;

  Select max(mark2) from result;

2. Table : Employee : emp_no (Primary key), emp_name, city, mgr_no, job, sal, dept_no

- Query:-
- Display department wise average salary.
- Ans.
- Create table employee

    (emp_no number(4) primary key,

    Emp_name varchar2(30),

    Mgr_no number(5),

    Job varchar2(20),

    Sal number(5),

    Dept_no number(5));

  Select dept_no, avg(sal) from employee group by dept_no;

Q3-A

- 1. A view is virtual table that can be accessed via SQL command.(true/false) Ans.True
- 2. In a __________ two database operations wait for each other to release a lock. Ans. deadlock
- 3.Oracle uses a method called ____________ to implement concurrency control.

  Ans. Multi-version concurrency control
- 4.Indexes may be created or dropped at any time.(True False)

  Ans.True

Q3-B

1. Differentiate : pessimistic lock v/s optimistic lock.

pessimistic lock	optimistic lock.
• Pessimistic locking is when you take an exclusive lock so that no one else can start modifying the record.	• Optimistic locking is when you check if the record was updated by someone else before you commit the transaction.
• Pessimistic locking aims to avoid conflicts by using locking.	• Optimistic Locking allows a conflict to occur, but it needs to detect it at write time.

2. Define the term : cluster , synonym.

- **Cluster :-**
- Clusters in SQL are used to store data that is from different tables in the same physical data blocks.
- They are used if records from those tables are frequently queried together.
- By storing same data blocks, the number of database block reads needed to full fill such queries decreases which improves performance.
- Each cluster stores tables data and maintains a clustered index to sort data.
- Columns within the cluster index are called clustered keys. These determine the physical placement of rows within the cluster.
- Cluster key is usually a foreign key of one table that references the primary key of another table in cluster.
- **Synonym :-**
- A synonym is an alternative name for objects such as tables, views, sequences, stored procedures, and other database objects.
- Synonyms is dependent object.

Q3-C

1. What is view ? Explain in detail.

- A view is a virtual table that stores only definition in the form of SQL statement. It is used to hide information from other user.
- **Feature of view : -**
- Creating view does not take any storage space in oracle
- The maximum number of columns that can be defined in a view is 1000
- When new records are inserted/updated/deleted in base table, the changes are automatically reflected in the view
- View once defined, can be used in Joins and Sub-query also
- **Types of view :-**
- Read only view
- Updateable view
- **Syntax :-**
- CRETE [OR REPLACE] [FORCE] VIEW <View Name> AS

  SELECT <Fields> FROM <Table> WHER <condition>

2. Write a note on indexes.

- An index is a object that contains an entry for each value that appears in the indexed column(s) of the table and provides direct, fast access to rows.
- **B tree indexes :-**
- B-tree is a self-balancing tree data structure that keeps data sorted and allows searches, sequential access, insertions, and deletions in time.
- Each page in an index B-tree is called an index node. The top node of the B-tree is called the root node. The bottom nodes in the index are called the leaf nodes.
- **Bitmap indexes :-**
- A bitmap is a mapping from one system such as integers to bits. It is also known as bitmap index or a bit array.
- The memory is divided into units for bitmap.
- These units may range from a few bytes to several kilobytes.
- Each memory unit is associated with a bit in the bitmap.

- If the unit is occupied, the bit is 1 and if it is empty, the bit is zero.
- The bitmap provides a relatively easy way to keep track of memory as the size of the bitmap is only dependent on the size of the memory and the size of the units.
- **Functional based index :-**
- Function-based indexes allow you to create an index based on a function or expression.
- Function-based indexes can involve multiple columns, arithmetic expressions, or maybe a PL/SQL function
- The following example shows how to create a function-based index:

  Create index emp_idx on emp (upper(ename));
- **Application Domain Indexes:-**
- Application domain indexes are custom indexes that are specific to the needs of your application.
- They can be based on the semantics ofyour data and queries.
- **Example:**
- You could create an application domain index like this:

  CREATE INDEX idx_library_availability ON books (branch_id, publication_date);

Q3-D

1. Write a note on sequence.

- A sequence is a set of integers that are generated in order on demand.
- Sequences are frequently used in databases because many applications require each row in a table to contain a unique value and sequences provide an easy way to generate them.
- Auto increment created in sequence so no need to remember field and insert data in that field with unique number.
- Sequences can be created, changed, and dropped with the following three SQL commands:
- CREATE SEQUENCE
- ALTER SEQUENCE
- DROP SEQUENCE

- Syntax :-
- Create sequence <name>

  [Start with <no>

  Increment by <no>

  Minvalue <no> / Nominvalue

  Maxvalue <no> / Nomaxvalue

  Cycle / Nocycle

  Cache / Nocache]
- A sequence definition may consist of a start value, increment value, minimum value, and maximum value.
- You can specify whether the sequence generator should stop when reaching a boundary value, or CYCLE the sequence numbers within the minimum/maximum range.
- All sequence attributes are optional and they all have default values.
- Each sequence has two columns: NEXTVAL and CURRVAL.
- NEXTVAL is use for nextvalue in sequence while CURRVAL use for current value of data.

2. What are locks? .Explain its various types.

- In Oracle PL/SQL, a LOCK is a mechanism that prevents destructive interaction between two simultaneous transactions or sessions trying to access the same database object.
- **Modes of Locking:-**
- Exclusive lock
- The mode prevents the associates resource from being shared .
- Share lock
- The mode allows the associated resource to be shared, depending on the operations involved .
- **Locking Issues:-**
- Lost Updates
- Pessimistic Locking
- Optimistic Locking
- Blocking

- Deadlocks
- Lock Escalation
- **Lost Updates :-**
- A lost update problem occurs due to the update of the same record by two different transactions at the same time.
- In simple words, when two transactions are updating the same record at the same time in a DBMS then a lost update problem occurs.
- **Optimistic and Pessimistic lock :-**
- Optimistic locking is when you check if the record was updated by someone else before you commit the transaction.
- Pessimistic locking is when you take an exclusive lock so that no one else can start modifying the record.
- **Blocking issue :-**
- Blocking issues in Oracle occur when a session holds an exclusive lock on an object and doesn't release it before another session wants to update the same data.
- This blocks the second session until the first session has done its work.
- Blocking locks can happen when a session issues an insert, update, or delete command that changes a row.
- When the change occurs, the row is locked until the session either commits the change or rolls the change back.
- **Deadlocks :-**
- A deadlock occurs when two or more sessions are waiting for data locked by each other, resulting in all the sessions being blocked.
- Oracle automatically detects and resolves deadlocks by rolling back the statement associated with the transaction that detects the deadlock.
- **Lock escalation :-**
- Lock escalation is a process that databases use to reduce the number of locks they hold.
- It occurs when a database raises the locks to a higher level on table.
- For example, if a single user locks many rows in a table, some databases automatically escalate the user's row locks to a single table.

- **Types of Locks :-**

Lock	Description
• DML locks (data locks)	• DML locks protect data • For example, table locks lock entire tables, rowlocks lock selected rows.
• DDL locks (dictionary locks)	• DDL locks protect the structure of schema objects
• Internal locks and latches	• Internal locks and latches protect internal database structures such as datafiles

Q4-A

- 1. cursor is a database object which is used to manipulate data in a row-to-row manner.
- 2. If user defined error condition exists, the call to the user defined exception is made using a raise statement.
- 3. Which is a database object that allows us to execute a batch of SQL code when a table event occurs?

  Ans. Trigger
- 4. Types data type is also known as user defined data type.

Q4-B

1. Differentiate: %type v/s %rowtype.

%type	%rowtype
• Used to declare a filed with the same type as • That of a specified table's column.	• Used to declare a record with the same types As found in the specified table, view or cursor.
• %TYPE is used to declare a variable or parameter whose data type is the same as a specific database column or variable.	• %ROWTYPE is used to declare a composite variable or parameter that can store an entire row of data from a table or cursor.
• You specify the source of the data type by referencing a column, variable, or another variable using the %TYPE attribute.	• You specify the source of the structure by referencing a table, cursor, or record using the %ROWTYPE attribute.

2. Define the term:- varrays, nested table.

- **Varrays :-** The varray (variable size array) is one of the three types of collections in PL/SQL (associative array, nested table, varray).
- A varray is similar to a nested table except you must specifiy an upper bound in the declaration
- **Nested Table :-** Nested tables are similar to PL/SQL tables, but they can be stored in the database as a column in a table or as a database object.
- Nested tables can be considered as unbounded arrays.

Q4-C

1. What is cursor? Explain its various types.

- When you execute a SQL statement from PL/SQL, the ORACLE assigns a private work area for that SQL statement is known as CURSOR.
- The PL/SQL is mechanism by which you can name that work area and manipulate the information within it. Data stored in the cursor known as Active Data Set (ADS).
- Types of cursors in oracle

  There are two types of cursor available in oracle

  1). Implicit cursor

  2). Explicit cursor
- **Implicit cursor :-**
- The implicit cursors are automatically generated by Oracle while an SQL statement is executed.
- These are created by default to process the statements when DML statements like INSERT, UPDATE, DELETE etc. are executed.
- **Attributes of both cursor :-**
- %ISOPEN – returns true if cursor open, false otherwise.
- %FOUND – returns true if record was fetched successfully, false otherwise.
- %NOTFOUND – returns true if record was not fetched successfully, false otherwise.

- %ROWCOUNT – returns number of records processed from the cursor.
- **Explicit cursor :-**
- A cursor can also be opened for processing data through a PL/SQL block such a user-defined cursor known as explicit cursor.
- Attributes of both cursor.
- %ISOPEN – returns true if cursor open, false otherwise.
- %FOUND – returns true if record was fetched successfully, false otherwise.
- %NOTFOUND – returns true if record was not fetched successfully, false otherwise.
- %ROWCOUNT – returns number of records processed from the cursor.

2. What is exception? Differentiate: predefined v/s user defined.

- PL/SQL supports programmers to catch such conditions using EXCEPTION block in the program and an appropriate action is taken against the error condition.

  There are two types of exceptions:-

Predefined	User defined
• Predefined data types, also known as built-in or system-defined data types, are provided by the database management system (DBMS) itself.	• User-defined data types are created by users or developers to meet specific requirements that cannot be fulfilled by predefined data types alone.
• These data types are part of the SQL standard or are specific to the particular DBMS you are using (e.g., SQL Server, Oracle, MySQL).	• These data types are defined and customized by users within the SQL database using CREATE TYPE or similar SQL statements.
• Common examples of predefined data types include INTEGER, VARCHAR, DATE, and BOOLEAN, among others.	• User-defined data types can be structured types (e.g., user-defined composite types, arrays, or object types) or non-structured types (e.g., user-defined domains or synonyms).

• These data types are readily available for use in your database schema without any additional definition or customization.	• They are typically used to abstract and encapsulate complex data structures or to enforce data integrity constraints.
• Example :- No_data_found	• Example:- when not_eligible then

Q4-D

1. What is procedure ? Explain with example.

- PL/SQL block is not stored in database, its compiled every time when its executed.
- To remove these restrictions, Oracle provides facility to declare 'Name Block' like Procedures, Functions, Packages and Triggers.
- The named block can be stored in the database and can be called and executed by client applications at any time.
- Procedures and functions are subprograms which store your PL/SQL block into database server of Oracle and executes when it needed.
- Procedures take zero or more parameters and return no values.
- Procedures are called as stand alone executable statements.
- **Perameter in procedure :-**
- You may pick one of the following modes for each parameter:
- IN is the default mode for a parameter.
- IN parameters already have a value when the procedure is run.
- The value of IN parameters may not be changed in the body.
- OUT is specified for parameters whose values are only set in the body.
- IN OUT parameters may already have a value when the procedure is called, but their value may also be changed in the body.
- **Syntax for procedure:**
- CREATE OR REPLACE PROCEDURE <procedure name>

  (<parameter list>) [IS/AS]

  <Declaration Section>

  BEGIN

  <PL/SQL code>

  EXCEPTION

<PL/SQL code>

END;

- Example :-
- create or replace procedure revstr (st in varchar2)as

```
    lst number;
    st1 varchar2(20);
begin
    lst:=length(st);
    for i in reverse 1.. lst
            loop
                    st1 := st1||substr(st,i,1);
            end loop;
    dbms_output.Put_line('Reverse of string is '|| st1);
end;
```

- OUTPUT :- SQL> exec revstr ('DAKU');

  Reverse Of String Is UKAD

2. Consider a student table having fields like seat.no, name, and mark. Develop a trigger on the table for insert or update any records and name should be display into the uppercase.

- -- Create the student table
- CREATE TABLE student (

```
    seat_no varchar2(25) PRIMARY KEY,
    name varchar2(30),
    mark int(3)
);
```

- -- Create the trigger
- CREATE TRIGGER uppercase_name

```
BEFORE INSERT ON student
FOR EACH ROW
BEGIN
```

Insert into student

Values

(:new.seat_no,upper(:new.name),:new.mark);

END;

- -- Trigger for updates
- CREATE TRIGGER uppercase_name_update

BEFORE UPDATE ON student

FOR EACH ROW

BEGIN

UPDATE student

SET name = upper(:new.name)

WHERE seat_no = :new.seat_no;

END;

Q5-A

- 1. DataFile File holds the actual data.
- 2. A table IS used to logically group data together.
- 3. LGWR stands for Log Writer Process .
- 4. SQL Server IS a principal scheduler object.

Q5-B

1. Differentiate: Data files v/s Redo log files.

Data files	Redo log files
• Data files store the actual data that is managed by the database. This includes tables, indexes, and other database objects.	• Redo log files serve as a critical component for ensuring the durability and recoverability of the database.
• When you perform INSERT, UPDATE, or DELETE operations on the database, the changes are made in the data files.	• Every time a transaction is committed, the changes made by that transaction are written to the redo log files.
• Data files are typically persisted on disk and are essential for the long-term storage and retrieval of data.	• Redo log files are cyclically overwritten, but their content is persistent until they are no longer needed for recovery purposes.

• In an e-commerce database, data files would store information such as customer details, product information, and order history.	• In case of a sudden power failure or system crash, redo log files can be used to replay the changes and bring the database back to its previous state.

2. Define the term: oracle blocks, import.

- **oracle blocks :** Oracle blocks are used to store various types of data within the database, including table data, index data, and other structural information.
- Each Oracle block has a specific size, which can vary depending on the configuration of the database, but it is typically 8KB (kilobytes).
- **Import :** the term "import" typically refers to the process of bringing data from an external source or file into a database.
- This is done to populate database tables with data from an external file or to update existing data within the database.

Q5-C

1. Write a note on initialization parameter.

- The initialization parameter file is a text file that contains a list of parameters and a value for each parameter.
- The file should be written in the client's default character set. Specify values in the parameter file which reflect your installation.
- The following are sample entries in a parameter file:

  PROCESSES = 100

  OPEN_LINKS = 12

  GLOBAL_NAMES = TRUE
- The name of the parameter file varies depending on the operating system. For example, it can be in mixed case or lowercase, or it can have a logical name or a variation on the name INIT.ORA.
- As the database administrator, you can choose a different filename for your parameter file. There is also an INITDW.ORA file which contains suggested parameter settings for data warehouses and data marts.
- Sample parameter files are provided on the Oracle Server distribution medium for each operating system.

- A distributed sample file is sufficient for initial use, but you will want to make changes in the file to tune the database system for best performance. Any changes will take effect the next time you completely shut down the instance and then restart it.
- **Database administrators can use initialization parameters to do the following:**

- optimize performance by adjusting memory structures, for example, the number of database buffers in memory
- set some database-wide defaults, for example, how much space is initially allocated for a context area when it is created
- set database limits, for example, the maximum number of database users
- specify names of files
- Many initialization parameters can be fine-tuned to improve database performance. Other parameters should never be altered or only be altered under the supervision of Oracle Corporation Worldwide Support staff.

2. What is memory structure ? Explain in detail.

- Memory Structures to the various data structures and components within a database management system (DBMS) that are used to manage and store data in memory.Below, I'll explain some of the key memory structures used in SQL databases in detail:
- Database buffer cache
- Redo log buffer
- Shared pool
- Dictionary cache
- **Database buffer cache:-**
- The database buffer cache is the portion of the SGA that holds copies of data blocks read from datafiles.
- The buffers in the cache are organized in two lists: the write list and the least recently used (LRU) list.
- The write list holds dirty buffers, which contain data that has been modified but has not yet been written to disk.
- The LRU list holds free buffers and dirty buffers that have not yet been moved to the write list.

- Free buffers do not contain any useful data and are available for use
- **Redo log buffer :-**
- The redo log buffer is a circular buffer in the SGA that holds information about changes made to the database.This information is stored in redo entries.
- Redo entries contain the information necessary to reconstruct changes made to the database by INSERT , UPDATE , DELETE ,CREATE , ALTER, or DROP operations.
- Redo entries are used for database recovery, if necessary.
- **Shared pool :-**
- Shared pool has two cache
- **Library cache**
- Executable code which connected to instance kept in library cache.

  Ex. Sql or pl/sql statements
- **Dictionary cache :-**
- Dictionary cache is stored library of all oracle statements like select, insert,update etc so when user enter any query dictionary cache checks its syntax in library

Q5-D

1. Write a note on scheduler.

- Oracle Scheduler Concepts refer to the fundamental ideas and components that form the basis of Oracle Scheduler, a powerful job scheduling and automation tool within the Oracle Database. Oracle Scheduler is designed to help automate and manage a wide range of tasks, jobs, and processes within the database environment.
- **Job:**
- A job is a specific task or unit of work that you want to automate using Oracle Scheduler. Jobs can include activities like running SQL scripts, PL/SQL procedures, external scripts, or data transformations.
- **Program:**
- A program defines what a job does. It specifies the executable and arguments required to perform a particular task. Programs can include stored procedures, executable files, or scripts.

- **Schedule:**
- A schedule defines when and how often a job should run. It specifies the frequency, start time, end time, and recurrence pattern for executing a job.
- **Job Class:**
- Job classes are used to categorize and manage jobs. You can assign jobs to specific job classes based on their characteristics or priorities.
- **Window:**
- A window is a specific time interval during which jobs are allowed to run. Windows are often used to control when maintenance tasks or resource-intensive jobs can be executed, ensuring they don't interfere with critical workloads.
- **Dependency:**
- Dependency defines the relationship between jobs. You can specify that one job should only run after another job has completed successfully. Dependencies enable you to create job chains and manage complex workflows.
- "Scheduling Jobs with Oracle Scheduler" involves using Oracle Scheduler, a feature within the Oracle Database, to automate and manage the execution of various tasks or jobs at specified times or under specific conditions.
- **Creating Jobs :**
- To schedule jobs with Oracle Scheduler, you first need to define the jobs you want to automate. Jobs can include tasks like running SQL scripts, PL/SQL procedures, or external scripts.
- **Defining Programs :**
- For each job, you need to specify a program that defines what the job does. The program specifies the executable or procedure to run, as well as any required arguments or parameters.
- **Setting Schedules :**
- Once you have defined jobs and programs, you need to create schedules that determine when and how often these jobs should run. Schedules can be configured to execute jobs at specific times of the day, week, or month, and they can include options like specifying start and end times or recurring patterns

- **Monitoring and Notifications:**
- Oracle Scheduler provides tools for monitoring the execution of scheduled jobs. You can view job status, logs, and performance metrics.
- **Job History:**
- Oracle Scheduler maintains a history of job executions, allowing administrators to track the progress of jobs, view job status, and diagnose problems.
- **Security and Access Control:**
- Access control mechanisms ensure that only authorized users can create, modify, or execute scheduled jobs.
- **Administering Oracle Schedule**
- **Job Management:**
- Administrators are responsible for creating, modifying, and deleting jobs. This includes defining the job type (e.g., SQL script, PL/SQL procedure), setting job attributes, specifying the program to execute, and assigning the job to schedules.
- **Programs and Executables:**
- Administrators manage programs and executables associated with jobs. This involves defining the programs, specifying executable files or PL/SQL procedures, and configuring program attributes.
- **Schedules:**
- Administering Oracle Scheduler involves the creation, modification, and deletion of schedules.
- **Dependency Management :**
- Administrators establish dependencies between jobs when needed. Dependencies define the order in which jobs are executed and ensure that specific conditions are met before a job starts.
- **Window Configuration :**
- Windows, which specify time intervals during which jobs are allowed to run, are set up and managed by administrators.
- **Resource Allocation :**
- Resource plans and job classes are configured to manage resource allocation to jobs.

2. What is instance architecture ? Explain.

- Instance Architecture the overall structure and components that make up a running instance of a DBMS.
- The term "instance" typically represents a single, independent copy of a database management system, which is capable of serving multiple users and processing database operations.
- When a database is started on a database server, Oracle allocates a memory area called the System Global Area (SGA) and starts one or more oracle processes.
- This combination of the SGA and the oracle processes is called an Oracle Instance
- **The instance and the Database :-**
- This is called Mounting the database. The database then ready to be Opened, which makes it accessible to authorized users.Only the database administrator can start up an instance and open the database.
- If a database is open, then the database administrator can shut down the database so that its closed
- Security for database startup and shutdown is controlled through connection to oracle with administrator privileges
- Normal users do not have control over the current status of an oracle database.
- **Instance and database startup :-**
- Start an instance
- Mount the database
- Open the database
- **Database and instance shutdown :-**
- Open the database
- Close the database
- Unmount the database
- Shut down the instance

- **An Oracle instance is build up by considering following aspects :-**
  1. Database Process
  2. Memory Structure
  3. Data Files
- **Database Process :-**
- The Oracle RDBMS uses two types of processes , user process and oracle process (called background process)
- **User processes :**
- Its user's connection to the RDBMS system. It communicates user input with the Oracle server process-by using oracle program interface.
- Its also used to display the information requested by the user.
- **Oracle processes :-**
- It performs the function for users which divided into two processes :
- A. server processes
- B. Background processes
- **Server process:-**
- Oracle creates server process to handle the request of user processes. This server creates for the for the following
- Run sql statements issued through the application
- Read necessary data blocks from data files on disk
- Return the result in such a way that the application can process the information
- **Background process :-**
- To maximize performance and accommodate many users, a multiprocess Oracle system uses some additional oracle processes called background processes.
- Background process in Oracle are follow :

➤ **Database Writer Process (DBW)**

➤ **Log Writer Process (LGWR)**

➤ **Checkpoint Process (CKPT)**

➤ **System Monitor Process (SMON)**

➤ **Process Monitor Process (PMON)**

➤ **Archiver Processes (ARCn)**

➤ **Recoverer Process (RECO**

➤ **Dispatcher**

➤ **Server**